

The Reporting of Greenhouse Gas Emissions: Market-based Scope 2 Emissions and Renewable Energy Instruments

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Abstract

We examine whether the current protocol for reporting Scope 2 greenhouse gas emissions motivates firms to direct their decarbonization efforts toward the use of market-based instruments to “buy-down” their environmental impact as opposed to engaging in real mitigation activities that offer greater credibility and impact in the context of global efforts to decarbonize. To do so, we conduct a difference-in-differences analysis surrounding the release of Scope 2 guidance by the Greenhouse Gas Protocol in 2015, which introduced the dual reporting of both a location-based and a market-based method for calculating emissions. Our findings suggest that the introduction of the market-based approach (which allows the purchase of renewable energy instruments to reduce reported emissions) prompted firms to change the nature of their mitigation efforts. In particular, our evidence indicates that market-based reporting motivated firms to choose cost-efficient means to achieve emissions targets at the expense of changes in real behavior that create additionality—i.e., the creation of policies or other interventions that result in emissions reductions over and above what would have occurred independently. We further show that this shift is stronger among firms with active (and more ambitious) Scope 2 emissions targets, consistent with targets increasing the demand for reported reductions when near-term physical abatement is costly. Overall, our evidence highlights how emissions reporting protocols can shape firms’ efforts to achieve emission-reduction goals, with direct implications for ongoing Scope 2 standard-setting debates as well as the mapping of firms’ contractual claims to their real-world emissions impact.

Keywords: carbon emissions, carbon accounting, greenhouse gas, additionality, decarbonization targets, renewable energy certificates, GHG Protocol

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1. Introduction

The measurement and reporting of greenhouse gas (GHG) emissions is fundamental to tracking progress in global efforts to mitigate climate change via decarbonization. Proponents of the current protocol for reporting Scope 2 (indirect) GHG emissions emphasize the need for a market-based approach that offers flexibility in measuring progress, noting that differences in business models, geographic locations, and operational barriers require alternative approaches to reducing emissions (Bryan et al., 2024; O’Brien, 2025). Critics argue that this flexibility compromises the accuracy and relevance of reported emissions and, as a result, undermines global mitigation efforts. In particular, the current market-based method of reporting allows companies to meet energy targets using contractual agreements that “report emissions numbers that bear little relation to their real-world pollution” (Brander et al., 2018; Bryan et al., 2024; Langer et al., 2024). Indeed, an oversupply of renewable energy instruments linked to clean energy generated in different grids at alternative times of day keeps prices very low.¹ This paves the way for some of the biggest energy users to purchase very cheap renewable energy certificates (RECs) to “buy-down their environmental impact” in order to reach emission reduction targets without regard for whether these efforts have any real impact on renewable energy supply (O’Neil, C., 2023; Brander et al., 2018; Bryan et al., 2024).

Consistent with these concerns, recent evidence indicates that firms purchase RECs to claim Scope 2 emission reductions but that RECs do not meaningfully reduce emissions or increase renewable energy generation (Bjorn et al., 2022; Langer et al., 2024; Bjorn et al., 2025). Consequently, a robust debate surrounds how companies should account for GHG emissions and the extent to which current reporting methods inflate estimates of emissions reductions and, in so

¹ For example, most unbundled RECs in the U.S. market traded for \$1–\$15 per megawatt hour in 2025 (SolarTech, 2025), compared with an average U.S. industrial electricity rate of approximately \$80 per megawatt hour (U.S. Energy Information Administration, 2025).

doing, hinder genuine progress in reducing global energy consumption. Moreover, rapid increases in Scope 2 emissions stemming from energy-intensive artificial intelligence (AI) models have prompted intensive lobbying efforts by tech firms to influence upcoming revisions to the current reporting guidelines (O'Brien, 2025). At the heart of this debate is the question of whether the current market-based approach to reporting emissions allows firms to choose cost-efficient means to reduce emissions at the expense of instituting changes in real behavior that create additionality—i.e., the creation of policies, projects, or other interventions that result in emissions reductions over and above what would have occurred independently.

To inform this discussion, we examine changes in firms' mitigation activities in response to the 2015 introduction of the market-based approach to accounting for Scope 2 GHG emissions. This allows us to provide timely evidence that speaks to a key open question as policymakers seek feedback on proposed updates to existing guidance: Does the current market-based method for reporting Scope 2 GHG emissions motivate firms to buy-down emissions as opposed to engage in real mitigation activities that more credibly support additionality?

In 2001, the Greenhouse Gas Protocol released the Corporate Accounting and Reporting Standard (Corporate Standard), which introduced a common framework for tracking corporate environmental performance.² Designed to be policy neutral with worldwide applicability in mind, the Corporate Standard became the globally recognized approach to reporting greenhouse gas emissions.³ A 2015 update to the Corporate Standard included guidance for Scope 2 emissions. This update required the dual reporting of location-based and market-based emissions with the aim of allowing each to reflect different risks and opportunities.

² See <https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf>.

³ The European Commission now mandates the GHG Protocol for sustainability reporting (European Commission, 2023).

Location-based reporting captures companies’ actual usage based on average emissions intensity from the local power grid. Market-based reporting reflects the extent to which the company’s energy purchasing decisions (via company-specific contractual agreements) reduce emissions. This means that changes in real mitigation behavior—in the form of reduced energy use or increased on-site renewable energy generation—drive changes in location-based emissions. In contrast, changes in market-based emissions can reflect changes in the firm’s real mitigation activities or, alternatively, can reflect the firm’s use of renewable energy instruments to achieve emission reduction targets. As such, the introduction of market-based emissions reporting in 2015 potentially incentivized the strategic (and cost-efficient) purchase of renewable energy instruments to achieve emission reduction targets.

To test this prediction, we conduct a difference-in-differences analysis surrounding the introduction of Scope 2 guidance in 2015. To do so, we assemble a sample of 7,532 firm-year observations for 1,078 unique companies operating in 35 countries during 2012 through 2021. Because we anticipate that mitigation efforts concentrate in firms with high levels of emissions, we identify treatment (control) firms as “high” (“low”) emitters with above- (equal to or below-) industry-median Scope 2 emission intensity in 2014.⁴ Consistent with a reporting-driven change in mitigation behavior, we detect significantly more purchases of renewable energy instruments for high emitters relative to low emitters beginning in 2016 and increasing in magnitude in subsequent years.

Further analysis reveals that the treatment effect concentrates in purchases of *unbundled* RECs, providing support for concerns about additionality. Unbundled instruments allow the buyer to consume regular grid power while purchasing certificates on the renewable generation outside of the local power grid and at different points in time. Because the renewable energy attribute (i.e.,

⁴ If low emitters also respond to Scope 2 guidance, this reduces our ability to detect treatment effects for high emitters.

the REC⁵) is sold separately from the electricity, this weakens the physical and temporal link to actual electricity consumption. This disconnect increases the supply of unbundled instruments, creating a low cost and flexible option to reduce reported energy consumption. In fact, companies can choose where, when, and what type of renewable energy source to support without altering their actual energy use. Yet, the benefits of unbundled instruments come at the cost of guaranteeing additionality.

In contrast, bundled renewable energy instruments involve the purchase of renewable electricity that can be directly tied to a specific source, such as a wind or solar project; the electricity and its renewable attribute are sold together, creating a strong link between energy use and renewable generation. While this bundled approach is commonly used in power purchase agreement (PPAs) and is generally more expensive, it offers greater credibility and impact in the context of efforts to reduce global energy consumption.

The evidence thus far suggests that the introduction of market-based reporting prompted high emitters to buy-down their environmental impact by taking advantage of the flexibility and cost efficiency offered by the global market for unbundled RECs. Our next set of tests explores this behavior in the context of firms' decarbonization targets.

A growing share of firms have voluntarily set decarbonization targets, with approximately 45% of U.S. companies disclosing target dates for net-zero commitments as of 2024 (MacFarland, Trapletti, and Fryer, 2024; MSCI Institute, 2024). Although the recent political climate and policy uncertainty have caused many U.S. firms to reduce the visibility of their decarbonization efforts, most continue to make progress toward their climate commitments (Hawkins and Cooper, 2025; PricewaterhouseCoopers, 2025; Segal, 2025). During our sample period we note that 38% of firm-

⁵ A Guarantee of Origin (GO) is used in Europe, while a REC is used in the U.S. Both act as tracking instruments to verify green energy consumption.

years (47% of firm-years after 2020) have active Scope 2 emissions targets, typically expressed as a specific emission reduction goal to be met by a target year in relation to a base year.

Considering the presence of these commitments, we find that firms with active targets drive the observed post-2015 spikes in purchases of unbundled renewable energy instruments. Subsequent tests indicate that these purchases are most pronounced for firms setting more aggressive targets, measured in terms of setting absolute reduction targets and in relation to reduction targets set by industry peers. Combined, this evidence suggests that firms view the purchase of renewable energy instruments as a tool in their quest to set and meet ambitious targets. Of course, the setting of targets signals these firms' commitments to decarbonization efforts. Thus, even absent the emergence of market-based reporting, we would expect to detect increased mitigation efforts for these firms. Our evidence suggests that the flexibility and cost savings associated with the market-based method for reporting emissions motivated these firms to direct their mitigation efforts toward the use of unbundled RECs to buy-down their environmental impact as opposed to toward real mitigation activities that more credibly support claims of additionality.

Treatment firms may increase their purchases of unbundled RECs as part of a more holistic transition to renewable energy. To explore this possibility, we examine measures of renewable energy consumption. We detect no statistically significant trend in treatment firms' consumption of renewable energy over time. In fact, in relation to control firms, we detect lower investment in renewable energy. We also examine treatment firms' decarbonization efforts. Although we find evidence of a link between the purchase of *bundled* renewable energy and reduced non-renewable ("dirty") energy consumption, we find no evidence of a similar link between *unbundled* RECs and energy consumption. Overall, this evidence is inconsistent with the notion that the rise in the purchase of unbundled RECs in the post-period accompanies treatment firms' transition to renewable energy or reflects broader efforts to decarbonize. Instead, our findings suggest more

isolated changes in mitigation behavior that reflect firms' efforts to use unbundled RECs as cost-efficient means to "buy down" their emissions and to meet decarbonization targets.

Additional analyses aim to address concerns stemming from confounding events, the potential for firms to misreport GHG emissions, and systematic differences between treatment and control firms. First, we examine dynamic treatment effects. In so doing, we provide evidence to support the parallel trends assumption. In terms of confounding events, we detect significant changes in treatment firms' reliance on RECs beginning in 2016. These changes predate the adoption of the Paris Agreement in November 2016 and the implementation of the European Union's Non-Financial Disclosure Directive (NFRD) in 2017. Second, we detect no significant changes in Scope 1 emissions when comparing high emitters of Scope 2 emissions to low emitters. This falsification test contradicts the notion that alternative events that caused firms to increase their mitigation efforts more comprehensively explain the isolated uptick we detect in the use of unbundled RECs following the release of Scope 2 guidance (and the market-based reporting method) in 2015. Third, with respect to the accuracy of firms' disclosures, recent evidence calls into question the reliability of self-reported emissions (Cohen, Rouen, and Sachdeva, 2026) but indicates that external assurance improves reporting quality (Gipper, Sequeira, and Shi, 2025). Limiting attention to a sample with assured GHG emissions, our results remain robust.⁶ Finally, to address systematic differences between treatment and control firms, we document the robustness of our results to various approaches to considering differences in firm characteristics (i.e., the inclusion of firm characteristics interacted with post-treatment indicators as well as entropy-balancing techniques).

⁶ The observability of actual energy purchased from the electricity grid as well as the purchase of renewable energy certificates facilitate the assurance of Scope 2 emissions. In contrast, assuring Scope 3 emissions is more involved, as it involves considering activities beyond the immediate control of the company.

Although a growing body of evidence questions whether RECs reduce carbon emissions (Langer et al., 2024), the current market-based approach to reporting Scope 2 emissions does not differentiate between decarbonization efforts that credibly support additionality (via direct links between renewable attributes and energy consumption) and mitigation efforts that cannot guarantee incremental reductions. As a result, critics believe that the current protocol thwarts global efforts to address climate change.⁷ We inform this discussion by providing evidence of a reporting-driven change in mitigation behavior that reflects firms' use of low-cost and flexible purchase agreements to buy-down their environmental impact. Accordingly, our findings call attention to the role that market-based reporting plays in shaping firms' efforts to mitigate climate change.

Our study contributes to the growing literature on GHG emissions and carbon accounting. In the context of Scope 1 emissions, Kim, Li, and Wu (2025) provide evidence that firms use cheap, low-quality carbon offsets to strategically boost ESG ratings. In the context of Scope 3 emissions, Carter, Lee and Yu (2026) document changes in firms' supply chain decisions in response to the threat of mandated disclosure of Scope 3 emissions. We complement these findings by showing that the current accounting protocol for reporting Scope 2 emissions influences the nature of firms' decarbonization efforts. That is, firms respond to market-based reporting rules by increasing reliance on unbundled RECs, which lowers reported emissions at low cost without ensuring additional renewable generation. By linking changes in mitigation behavior to the reporting guidelines themselves, our evidence speaks directly to policymakers' request for feedback on Scope 2 reporting guidance as they currently deliberate updates to existing protocol.⁸

⁷ Professor Matthew Brander compiles evidence about additionality in the context of Scope 2 emissions reporting. See <https://www.bccas.business-school.ed.ac.uk/research/carbon-accounting-pricing-policy/renewable-energy-purchasing>.

⁸ See <https://ghgprotocol.org/sites/default/files/2025-10/GHG-Protocol-Scope2-Public-Consultation.pdf> and <https://kpmg.com/xx/en/our-insights/ifrg/2025/ghg-protocol-scope2-consultation.html>.

This evidence is particularly relevant given the surge in Scope 2 emissions from energy-intensive AI models, which has placed Scope 2 guidance at the center of ongoing debates about how tech companies account for data center emissions (O'Brien, 2025).

Investors demand compensation for exposure to carbon risk (Matsumura, Prakash, Vera-Munoz, 2014; Bolton and Kacperczyk, 2021), compel firms to disclose information about climate risks (Flammer, Toffel, and Viswanathan, 2021; Pawliczek, Skinner, and Wellman, 2021), and increasingly pressure firms to reduce their GHG emissions (Naaraayanan, Sachdeva, and Sharma, 2021; Safiullah, Alam, and Islam, 2022; Kahn, Matsusaka, and Shu, 2024). Yet, firms often fail to meet emissions-reduction goals (e.g., Ioannou, Li, and Serafeim, 2016; Dahlmann, Branicki, and Brammer, 2019) and investors typically do not punish firms when targets are missed (Jiang, Kim, and Lu, 2025; Bolton and Kacperczyk, 2025). Our findings offer one potential explanation for why investors might ignore target outcomes despite finding carbon emissions value-relevant as they indicate that skepticism is warranted when interpreting progress toward decarbonization. Firms direct their mitigation efforts toward the use of unbundled RECs in pursuit of ambitious targets rather than toward efforts that more credibly support additionality. Accordingly, evaluating progress toward decarbonization targets requires attention not only to whether firms report reductions, but also to the ways in which those reductions are achieved.

More broadly, our study adds to research documenting the real effects of accounting standards and reporting regimes on firms' economic behavior (Kanodia and Sapra, 2016). Prior work shows that changes in measurement and disclosure requirements alter incentives and contracting choices in settings such as pension reporting (Chuk, 2013; Anantharaman and Chuk, 2018) and lease accounting (Ma and Thomas, 2023; Li and Venkatachalam, 2024). With respect to carbon disclosures, Tomar (2023) finds that mandatory facility-level GHG emissions disclosures lead to emissions reductions, while Carter et al. (2026) document changes in firms' operations in anticipation of changes to mandated disclosure of Scope 3 emissions. Our study extends this

literature to a voluntary, nonfinancial reporting setting by showing that the *method of measurement*—not merely the act of disclosure—shapes firms' mitigation choices. By highlighting how the introduction of market-based reporting of Scope 2 emissions can redirect effort toward contractual purchase agreements rather than toward more direct mitigation activities aimed at reducing global energy consumption, our findings provide timely evidence that informs current policy deliberations.

2. Background and hypothesis development

2.1 Accounting for carbon emissions

The GHG Protocol provides the globally recognized standard for carbon accounting. Developed by the World Resources Institute (WRI) and the World Business Council for Sustainable Development (WBCSD) in 2001, the GHG Protocol establishes the framework for identifying, calculating, and reporting Scope 1 (direct), Scope 2 (indirect energy), and Scope 3 (value chain) emissions. Scope 1 emissions stem from sources directly controlled by the company; examples include emissions from equipment such as boilers, furnaces, vehicles, as well as emissions from production equipment. Scope 2 and Scope 3 emissions cover indirect sources of emissions. Scope 2 emissions stem from a company's purchase of energy, which includes electricity, steam, heating, and cooling; this includes grid electricity. Scope 3 emissions include all other indirect emissions stemming from a company's upstream and downstream emissions.

Issued via an update in 2015, the Scope 2 guidance represents the most significant change to the GHG Protocol since its initial release in 2001. In the decade since its 2015 issuance, controversy surrounding the requirement for the dual reporting of a location-based and a market-based method for calculating Scope 2 emissions has grown. While location-based reporting reflects the physical reality and geography of electricity generation, market-based reporting reflects the energy that companies obtain via supplier contracts, RECs, GOs, etc. As such, critics of the 2015

protocol argue that permitting the market-based approach to reporting emissions undermines genuine progress toward decarbonization by allowing companies to substitute the purchase of cheap certificates for changes in real behavior that more credibly support clean energy goals (O’Neil, C., 2023; Brander et al., 2018; Bryan et al., 2024).

In response to growing concerns, the GHG Protocol proposed changes to its standards in 2025, including major changes to its Scope 2 Guidance. With evaluation of public feedback currently in process, debate about the role that carbon-accounting methodology plays in shaping companies’ decarbonization efforts continues, including in the context of carbon emission reporting from data centers driven by the increasing use of AI (O’Brien, 2025) that use vast amounts of electricity.

In its current form, the 2015 Scope 2 Guidance standardizes the calculation of Scope 2 emissions by calling for the dual reporting of emissions using two separate methods: location-based and market-based. Actual energy usage based on average emissions intensity from the local power grid determines the location-based calculation. In contrast, market-based calculations adjust for the purchase of energy contracts. If a firm discloses its market-based emissions, it must also disclose its location-based emissions to be in compliance with the GHG Protocol’s Scope 2 Guidance. These distinct approaches and the required reporting of both types facilitate the comparison of differences in decarbonization efforts across firms and over time.

2.2 Disclosure of carbon emissions and reduction targets

The CDP (formerly Carbon Disclosure Project) serves as the primary source of firm-level carbon disclosures. Founded in 2000, the CDP operates as an international non-profit organization with the aim of facilitating environmental reporting via its collection of self-reported data from a broad cross-section of companies. As part of this mission, the CDP administers an annual, standardized questionnaire, which includes information about firms’ Scope 1, 2, and 3 emissions. Climate-conscious investors become signatories of the CDP to access these disclosures. This

investor-driven demand for climate information can induce portfolio firms' disclosures of climate information with the CDP (Cohen et al., 2023).

In conjunction with a commitment to disclosing their carbon emissions, a growing number of firms have voluntarily set decarbonization targets (Jiang et al., 2025; Bolton and Kacperczyk, 2025). Disclosed emission reduction targets typically consist of (i) a start year (i.e., the year when the reduction target is set), (ii) a target horizon (i.e., the difference between the target year and the base year, which is the year to which future emissions are compared), (iii) the target goal (i.e., the percentage of emissions that the firm aims to reduce), (iv) the target scope (i.e., Scope 1, market-based or location-based Scope 2, Scope 3 emissions, or a combination of any of these), and (v) the target type (either intensity or absolute target).

Because these targets are not binding, progress toward achieving them, as well as how decarbonization occurs, varies by firm. In fact, many firms fail to meet emission reduction targets, with prior work documenting missed (or quietly rescinded) targets for approximately 35 to 60 percent of firms (Ioannou et al., 2016; Bolton and Kacperczyk, 2025; Jiang et al., 2025). When firms do achieve decarbonization goals, this can reflect strategic efforts to set easily attainable benchmarks and/or the purchase of renewable energy instruments to offset energy consumption. At the same time, progress toward targets can also result from changes in firms' real mitigation behavior in the form of reduced energy usage or increased on-site renewable energy generation.

2.3 Predictions

Existing work finds that carbon emissions are negatively associated with firm value (Matsumura et al., 2014) but positively associated with future stock returns (Bolton and Kacperczyk, 2021), suggesting investors demand compensation for exposure to carbon risk. Meanwhile, studies also find that investors increasingly pressure firms to reduce their carbon footprints (Azar et al., 2021; Cohen et al., 2023). Collectively, this evidence supports the value-relevance of carbon emissions and, in fact, indicates that firms have an incentive to decarbonize.

We anticipate that the 2015 release of Scope 2 Guidance by the GHG Protocol changed how firms decarbonize. Prior studies find that changes in measurement and disclosure requirements alter firms' real activities in settings such as pension reporting (Chuk, 2013; Anantharaman and Chuk, 2018) and lease accounting (Ma and Thomas, 2023; Li and Venkatachalam, 2024). Consistent with this evidence, we expect that the introduction of market-based emissions, which allow for the purchase of renewable energy instruments to report lower emissions, prompted firms to change the nature of their mitigation activities. In particular, we predict that market-based reporting motivated firms to choose cost-efficient means to achieve emissions targets at the expense of changes in real behavior in the form of reduced energy or increased on-site renewable energy generation.

We further predict this shift in firms' mitigation efforts will concentrate in the purchase of *unbundled* renewable energy instruments. Bundled instruments ensure additionality, which offers greater credibility and impact in the context of efforts to reduce global energy consumption. Yet, their contracts have distinct disadvantages. Because the renewable attribute and the physical electricity it represents are generated and sold simultaneously (typically via a power purchase agreement, i.e, PPA), this increases the cost while also tying the firm to specific, long-term, and location-dependent logistical and contractual constraints. In contrast, unbundled RECs offer a cheap and flexible means to reduce reported emissions but do so at the cost of guaranteeing additionality.

3. Sample and research design

3.1 Sample

Our primary analyses rely on survey data on carbon emissions from the CDP, covering an international sample from 2012 to 2021. We merge the CDP data with the Compustat Global dataset to obtain firm-level financial characteristics. Panel A of Table 1 presents the distribution of

our sample firms by country. The U.S. is the most represented country with 1,946 firm-years, followed by Japan with 1,055 and the U.K. with 726 firm-years, respectively. The countries with the least representation are Poland (8 firm-years), Chile (13 firm-years), and Colombia (14 firm-years).

Panel B of Table 1 presents the industry distribution of our sample firms based on the Fama-French 12-industries classification. The largest proportion of our sample firms arises from manufacturing (15.91 percent) and business equipment (13.71 percent). The industries with the least representation in our sample are consumer durables (3.62 percent), oil, gas and coal extraction and production (3.70 percent), and finance (4.41 percent).⁹

3.2 Research design

In our main tests, we conduct the following difference-in-differences regression analysis:

$$Dep_{it} = \beta_0 + \beta_1 HighScope2_i \times Post_t + \gamma_1 X_{it} + \mu_i + \eta_{ct} + \varepsilon_{it}. \quad (1)$$

We identify *HighScope2* firms as those with above-median Scope 2 intensity within their industry in 2014.¹⁰ The variable *Post* is a time dummy that equals one for years 2015 and onward, the year the GHG Protocol released the Scope 2 Guidance that introduced a market-based method for calculating emissions. Our variable of interest is β_1 , which captures the differential reaction of high emitters versus low emitters to the release of the Scope 2 guidance.

The vector X_{it} contains time-varying firm-level controls including cash holdings; leverage; size; return on assets; property, plant, and equipment; as well as institutional ownership. Please see Appendix A for detailed variable definitions. We include firm fixed effects μ_i to absorb time-invariant, unobserved firm heterogeneity. We also include country-year fixed effects (η_{ct}) and industry-year fixed effects in some specifications (η_{jt}) to control for time-varying unobserved

⁹ Consistent with prior research in the carbon accounting literature (e.g., Ioannou et al., 2016), we include financial firms (SIC codes 6000–6999) in our analysis.

¹⁰ Differences in reporting behavior in the pre-period may introduce noise in our identification of treatment firms.

macroeconomic conditions that influence all firms in a specific country or all firms within a given industry, respectively. These fixed effects ensure that the results are not driven by economy-wide or industry-specific trends. Standard errors are clustered at the firm level.

Our main outcome variable of interest is renewable energy instruments (*RenEnergyInstr*), which firms purchase and employ to lower their market-based Scope 2 emissions. Depending upon the specification, the dependent variable takes different transformations of *RenEnergyInstr*. Appendix B presents the top sourcing methods of renewable energy instruments separately for unbundled purchases and bundled commitments of such instruments. Typical unbundled purchases are unbundled energy attribute certificates (EACs), RECs, and GOs. The top sources of bundled commitments include off-grid energy consumption from an on-site installation or through a direct line to an off-site generator and PPAs.

Panel C of Table 1 presents summary statistics for the main variables used in the regression analyses. We note that 67 percent of firm-years have some renewable energy instruments (*RenEnergyInstr_Dummy*). The inverse hyperbolic sine (IHS) transformation of renewable energy instruments scaled by revenue, $\sinh^{-1}(\text{RenEnergyInstr})$ reveals substantial variation, with a mean of 1.402 and a standard deviation of 1.881. $\sinh^{-1}(\text{BundledREI})$ and $\sinh^{-1}(\text{UnbundledREI})$ measure the amount of renewable energy instruments related to bundled commitments and unbundled purchases, respectively. The summary statistics reveal that, on average, about 2.5 times more purchased renewable energy pertains to unbundled purchases rather than bundled commitments. While just descriptive, these statistics indicate that firms are more likely to invest in unbundled purchases of renewable energy instead of committing to sustained efforts to lower Scope 2 emissions. The descriptives also reveal that firms seem to consider unbundled and bundled instruments somewhat as substitutes. The median of both these two variables is zero, while about 67 percent of firm-years have some type of renewable energy certificates (*RenEnergyInstr_Dummy*).

4. Results

4.1 Main findings

Table 2 presents the results for Equation (1) using different dependent variables and specifications, i.e., with and without the inclusion of control variables as well as with firm and country-year or firm and industry-year fixed effects, respectively. Columns (1) through (3) present regression results for the extensive margin with the dependent variable *RenEnergyInstr_Dummy*, which equals one if firms use any renewable energy instruments in the respective year, and zero otherwise. The interaction coefficient for *HighScope2×Post* is significantly positive for all specifications, ranging from 0.0523 to 0.0734. These results suggest that high emitters are about 5 to 7 percent more likely to use renewable energy instruments than low emitters after the release of the GHG Protocol's Scope 2 Guidance in 2015.

Next, Columns (4) through (6) present regression results for $\ln(RenEnergyInstr)$ as the dependent variable. This regression specification focuses on the intensive margin because the sample consists only of firm-years with any renewable energy instruments. Conditional on using any renewable energy instruments, high emitters use significantly more renewable energy instruments relative to low emitters in the post period. In terms of economic magnitude, the coefficients suggest that high emitters purchase about 72 to 125 percent more renewable energy than low emitters after the issuance of the Scope 2 Guidance.¹¹

Finally, Columns (7) through (9) present regression results for *RenEnergyInstr*, which equals the amount of purchased renewable energy and zero if no renewable energy instruments exist. Similar to Kim, Shi, and Verdi (2025), we use the IHS transformation $\sinh^{-1}(RenEnergyInstr)$, which addresses frequent zero values in the variable distribution.¹² The coefficients on *HighScope2×Post* remain significantly positive and range in magnitude from

¹¹ We calculate the percentage change as $(e^{\beta_1}-1) \times 100$.

¹² The remaining analyses focus on $\sinh^{-1}(RenEnergyInstr)$ for brevity.

0.3506 to 0.4027, suggesting that high emitters use around 35 to 40 percent more renewable energy instruments in the post period than low emitters.

As discussed in Subsection 2.3 and detailed in Appendix B, renewable energy instruments differ in their commitment levels. Specifically, certain renewable energy instruments are unbundled certificates while other instruments indicate firms' commitment to the bundled purchase of renewable energy, such as power purchase agreements (Environmental Protection Agency, 2024).

Table 3 presents results for Equation (1) separately for $\sinh^{-1}(UnbundledREI)$ and $\sinh^{-1}(BundledREI)$ in Columns (1) through (3) and Columns (4) through (6), respectively. The findings indicate that the treatment effect for high emitters concentrates in purchases of unbundled RECs. That is, high emitters purchase significantly more unbundled renewable energy instruments in the post period relative to low emitters, but the amount of bundled renewable energy instruments does not differ between the two groups in the post period. These results suggest that the Scope 2 guidance incentivizes firms to use unbundled renewable energy instruments to lower their market-based Scope 2 emissions instead of using bundled renewable energy instruments, which are more likely to indicate a firm's commitment to long-term decarbonization efforts.

Figure 1 presents dynamic effect coefficients for the interaction of *HighScope2* with year indicators using 2014 (the last year prior to the Scope 2 Guidance) as the base year for $\sinh^{-1}(RenEnergyInstr)$ and $\sinh^{-1}(UnbundledREI)$ as the dependent variables in Panels A and B, respectively. Both panels reveal no significant pre-trends in high emitters' purchases of renewable energy relative to low emitters. Starting with 2015, the year of the Scope 2 guidance issuance, we observe an increasing trend of treated firms' usage of total and unbundled renewable energy instruments. The treatment coefficient becomes significantly positive at the ten-percent level in 2016 for unbundled instruments and in 2017 for total instruments.

Overall, the findings in Tables 2 and 3 as well as Figure 1 paint a consistent picture that high emitters purchase more renewable energy, both at the extensive and intensive margin, after the GHG Protocol releases its Scope 2 guidance but that these purchases are concentrated in unbundled renewable energy certificates. Therefore, these results are consistent with the change of accounting treatment of renewable energy instruments having real effects on firms' purchases of such instruments.

4.2 Emission reduction targets

One reason why firms may purchase renewable energy instruments could be their establishment of voluntary Scope 2 emission reduction targets. That is, firms may purchase renewable energy instruments to meet their Scope 2 targets instead of conducting real decarbonization efforts. Accordingly, consistent with targets increasing the demand for reported reductions when near-term physical abatement is costly, we predict that post-2015 spikes in the use of contractual agreements to buy-down emissions will be strongest among firms with active and more ambitious Scope 2 emissions targets. Indeed, we expect that the flexibility of the market-based approach to accounting for carbon emissions caused firms to view the purchase of renewable energy instruments (particularly unbundled RECs) as a valuable tool in their quest to set and meet ambitious decarbonization targets.

Directing decarbonization efforts toward the use of unbundled RECs to buy-down their environmental impact instead of purchasing bundled renewable energy instruments (e.g., PPAs) may be the dominant strategy for firms that aim to respond to stakeholders' demands for lower emissions. Yet, this strategy assumes that stakeholders value reduced market-based emissions regardless of the method by which lower reported emissions are achieved. To the extent that firms respond to climate-conscious investors by engaging in comprehensive decarbonization efforts that more credibly support additionality, this predicts increased mitigation efforts more broadly over

time—as opposed to an isolated uptick in the purchase of unbundled RECs following the introduction of Scope 2 Guidance.

We test this prediction through a set of analyses based on firms’ target setting behavior. First, we test whether firms purchase more renewable energy certificates if they have active Scope 2 reduction targets.

Panel A of Table 4 presents triple difference-in-differences regression results including an indicator *Target*, which equals one if the respective firm-year has an active Scope 2 target, and zero otherwise.¹³ We augment Equation (1) to include the triple difference-in-differences coefficient $HighScope2 \times Post \times Target$ as well as $Target \times Post$, $Target \times HighScope2$, and $Target$. Columns (1) to (3) use the total amount of renewable energy instruments, $\sinh^{-1}(RenEnergyInstr)$, as the dependent variable. The results show that high emitters with active Scope 2 targets use about 2.2 to 2.5 times more renewable energy instruments than high emitters without active Scope 2 targets in the post period and relative to low emitters (i.e., $0.5213/0.2396 \approx 2.2$ and $0.4787/0.1942 \approx 2.5$). These results suggest that firms are more likely to purchase renewable energy instruments when firms have active Scope 2 targets. Columns (4) through (6) and (7) through (9) of Panel A repeat the analysis for unbundled renewable energy certificates and bundled renewable energy instruments, respectively. These analyses reveal that the results are concentrated in unbundled renewable energy certificates. That is, firms with active Scope 2 emissions targets are about 2 times more likely to purchase unbundled renewable energy instruments in the post period than high emitters without an active target.

Second, we expect that the purchase of renewable energy instruments differs based on the type of the Scope 2 reduction target. Specifically, absolute targets are likely more ambitious than intensity targets. Intensity targets generally scale the carbon emissions by revenue and, thus,

¹³ *Target* equals one in 37.7 percent of firm-years (Table 1 Panel C).

intensity targets may also be achieved if sales grow faster than carbon emissions. Panel B of Table 4 repeats the triple difference-in-differences regression from Panel A but splits the reduction targets by target type, i.e., absolute and intensity targets, in Columns (1) through (3) and (4) through (6), respectively.¹⁴ Because the existence of a Scope 2 target increases the purchase of unbundled renewable energy certificates but not bundled renewable energy instruments (Panel A of Table 4), Panel B focuses on unbundled renewable energy certificates as the dependent variable for conciseness.¹⁵ Consistent with absolute targets being more ambitious and, thus, firms having to rely more on renewable energy instruments to meet these reduction targets, we find that high emitters purchase around 3.25 to 4.10 times¹⁶ more unbundled renewable energy instruments in the post period relative to low emitters when they have active absolute targets relative to those without an absolute target. In contrast, we do not find that high emitters purchase more unbundled renewable energy certificates in the post period relative to low emitters when they have intensity targets.

Third, we conjecture that setting more ambitious targets may be a factor contributing to the purchase of renewable energy certificates. Thus, for the subsample of firm-years with active Scope 2 targets, we create an indicator variable *AmbitiousTarget*, which equals one if the firms' target reduction goal, measured as the planned carbon reduction over the target horizon¹⁷, is in the top quartile in the respective industry-year, and zero otherwise.¹⁸ Panel C of Table 4 presents the results of the triple difference-in-differences regression using *AmbitiousTarget* as the interaction variable. Columns (1) to (3) use $\sinh^{-1}(\text{RenEnergyInstr})$ as the dependent variable. The main

¹⁴ Approximately 25.4 and 17.8 percent of firm-years have absolute and intensity targets, respectively (Table 1 Panel C).

¹⁵ As in Panel A, untabulated results confirm that our results hold for the total amount of renewable energy certificates while we do not observe any significant changes in the purchase of bundled renewable energy instruments.

¹⁶ $0.8087/0.1984 = 4.08$, $0.7975/0.1946 = 4.10$, and $0.8297/0.2556 = 3.25$, respectively, for Columns (1) through (3).

¹⁷ For instance, if a firm sets a goal to reduce emissions by 20% over five years, its ambitiousness score would be 4% per year ($= 20\% \div 5$).

¹⁸ Panel C in Table 1 presents descriptives for the relative target reduction goal (planned carbon reductions relative to the target horizon), which averages 3.637.

coefficient of interest is the triple difference-in-differences coefficient $HighScope2 \times Post \times AmbitiousTarget$, which suggests that high emitters use about 1.8 to 2 times more renewable energy instruments in the post-guidance period relative to low emitters when they set ambitious targets. These results are concentrated in the purchase of unbundled renewable energy instruments. In summary, the findings in Panel C of Table 4 suggest that high emitters purchase more renewable energy instruments when their Scope 2 targets are more ambitious.

4.3 Additional analyses

4.3.1 Renewable energy transition

The purchase of unbundled renewable energy instruments may signal a firm's transition towards more renewable energy consumption. If this were the case, we would observe that high emitters consume more renewable energy relative to low emitters in the post period.

To examine this assertion, we regress the percentage of firms' renewable energy consumption on the indicator *HighScope2* plus control variables and country-year (plus industry-year) fixed effects. The sample period of this test begins in 2017, which is the first year of the CDP's collection of the renewable energy consumption data.

Columns (1) through (3) in Panel A of Table 5 present the results and show that *HighScope2* firms consume less renewable energy than low emitters.¹⁹ While these results reflect associations rather than causal estimates, they are inconsistent with treatment firms consuming more renewable energy and thus transitioning to more renewable energy consumption in the post-guidance period.

To further examine treatment firms' transition to renewable energy consumption, Columns (4) through (6) in Panel A of Table 5 interact *HighScope2* with a linear time variable *Time*, which captures the years relative to year 2017 (the first year of the sample in Table 5). If treatment firms were transitioning to more renewable energy over time, we would expect a significantly positive

¹⁹ Please note that *HighScope2* is defined based on the location-based Scope 2 emissions in 2014 and thus does not make a statement about the source of the energy consumption.

coefficient on this interaction variable. Inconsistent with this hypothesis, the interaction term remains statistically insignificant at conventional levels.

Finally, if high emitters' purchases of renewable energy instruments indicated a renewable energy transition, we expect a negative relationship between high emitters' purchases of renewable energy instruments and the consumption of non-renewable ("dirty") energy. Therefore, Panel B of Table 5 presents results for non-renewable energy consumption as the dependent variable and interacts *HighScope2* with an indicator for whether a firm has above-median total (Columns (1) and (2)), unbundled (Columns (3) and (4)), or bundled (Columns (5) and (6)) renewable energy instruments, respectively. The interaction coefficient on *HighScope2* with above-median total and unbundled renewable energy instruments is statistically insignificant. It only becomes significantly negative at conventional levels for bundled renewable energy instruments. These findings are inconsistent with above-median purchases of unbundled renewable energy instruments of high emitters being indicative of a renewable energy transition.

In summary, the findings in Table 5 are inconsistent with high emitters' renewable energy transition in the post-guidance period.

4.3.2 Subsample of assured Scope 2 emissions

One concern about the main results could be that the Scope 2 emissions are non-verified disclosures, which opens the possibility for misreporting. Given that the disclosure of such Scope 2 emissions is largely voluntary, we consider firms' intentional misreporting of these numbers unlikely.²⁰ Instead of intentional misreporting, and thus essentially committing fraud, firms can choose a non-disclosure regime. Non-disclosure should be less costly than intentional misreporting

²⁰ One exception is the EU, which mandated ESG disclosures, including carbon emissions, with the non-financial disclosure directive (NFRD) in 2014. We show robustness of our results to excluding the EU firms from the sample in Table 8, as discussed in this section below.

due to the non-zero risk of fraud detection and its resulting costly consequences, such as criminal charges, lawsuits, and loss in firm value.

To further eliminate the concern that firms misreport their Scope 2 emissions, we restrict the sample to firm-years with assured Scope 2 emissions, which minimizes the misreporting concern. Table 6 presents the results for the assurance sample and confirms our main findings that high emitters purchase significantly more renewable energy instruments in the post-Scope 2 guidance period relative to low emitters. As in the main analyses, these results are concentrated in the purchase of unbundled renewable energy certificates. Therefore, our results are unlikely to be driven by misreported Scope 2 disclosures.

4.3.3 Other macroeconomic developments

Another concern is that firms change their behavior around the issuance of the Scope 2 guidance due to other macroeconomic events. For instance, the RE100 is a global initiative, launched during the 2014 Climate Week, with the RE100 corporate members' mission to achieve a renewable energy transition.²¹ While the launch of the RE100 initiative coincides with the issuance of the Scope 2 guidance, by 2015 fewer than 50 firms had joined the initiative globally. Therefore, it is unlikely that our results can be explained by this initiative. Relatedly, the Paris Agreement was adopted by 196 countries in December 2015.²² The agreement became binding by November 2016. While the Paris Agreement sets goals for participating countries, translating these goals into firm-level effects requires time for domestic regulatory implementation. Such an effect would likely not occur before 2017 (or later), because countries need some time to take regulatory actions adopting the goals of the Paris Agreement. As shown in Figure 1, we find an impact on firms' usage of renewable energy instruments already in 2016, which predates the most plausible timeline for firm-level effects of the Paris Agreement.

²¹ <https://www.there100.org/about-us>.

²² <https://unfccc.int/process-and-meetings/the-paris-agreement>.

We further test whether our findings could be attributed to the Paris Agreement by conducting a falsification test using Scope 1 emissions intensity as the dependent variable in Equation (1). Because the Paris Agreement does not specifically target Scope 2 emissions, which are the focus of our study, we would expect to also observe changes in firms' Scope 1 greenhouse gas emissions if our findings were attributable to the Paris Agreement and not the GHG Protocol's issuance of the Scope 2 guidance. Inconsistent with this expectation, we do not find any significant differences in Scope 1 intensity for treatment firms relative to control firms in the post-guidance period.

In 2014, the European Union (EU) adopted the Non-Financial Reporting Directive (NFRD), which requires firms to prepare annual sustainability reports, including disclosures on greenhouse gas emissions, starting in 2017. Although Panel B of Figure 1 shows significant treatment trends already in 2016, we repeat our main analyses excluding firms headquartered in EU countries. Inconsistent with our results being driven by the adoption of NFRD, our results are robust to the exclusion of EU firms (Table 8).

Overall, the results in Tables 7 and 8 bolster our interpretation that the introduction of Scope 2 guidance—instead of some other macroeconomic shocks, such as the Paris Agreement or NFRD—explains our results.

4.3.4 Disclosure gap

If high emitters use more renewable energy certificates in the post-guidance period, we expect to observe a larger disclosure gap (i.e., difference) between market-based and location-based emissions for high emitters relative to low emitters. Because dual reporting was introduced with the Scope 2 guidance, this test is an association test in the post-guidance period. We regress *DisclosureGap* on the indicator *HighScope2* plus control variables and fixed effects in the sample period from 2015 to 2021. Consistent with expectations, Table 9 shows that treatment firms have a significantly higher disclosure gap than control firms in the post-guidance period. These results

provide corroborating evidence that treatment firms rely on renewable energy instruments to reduce their market-based emissions relative to their actual location-based emissions.

4.3.5 Additional robustness tests

Because we identify treatment firms based on the median Scope 2 emissions in the firm's respective industry, industry differences correlated with their reliance on Scope 2 emissions cannot explain our results. Yet, within an industry, treatment firms may differ from control firms along firm fundamentals. To address this concern, Panel A of Table 10 expands on Equation (1) by interacting the firm-level control variables with the post indicator. If any of these observable firm characteristics explained our baseline results, the main coefficient of interest, *HighScope2*×*Post*, would lose significance. Inconsistent with firm fundamentals explaining the observed post-treatment effect, *HighScope2*×*Post* remains significantly positive at a one percent significance level.

In a second robustness test to address the concern about differences across firm fundamentals, we entropy balance treatment and control firms in 2014 (the year prior to the Scope 2 guidance issuance) on their first and second moments, ensuring that treatment and control firms do not differ significantly in means and variances across the control variables. Panel B of Table 10 shows that our main results are robust to this alternative research design and suggest that treatment firms use about 56 percent more renewable energy instruments in the post-guidance period relative to control firms. In a final robustness test, we run a Poisson regression. Panel C of Table 10 shows that our results are robust to this alternative regression specification.

5. Conclusion

We examine whether the current protocol for reporting Scope 2 greenhouse gas emissions motivates firms to direct their decarbonization efforts toward the use of market-based instruments to buy-down their environmental impact. Our findings suggest that the introduction of the market-

based approach (which allows the purchase of renewable energy instruments to reduce reported emissions) prompted firms to change the nature of their mitigation efforts. Indeed, we find that market-based reporting motivates firms to choose cost-efficient means to achieve emissions targets at the expense of changes in real behavior that create additionality—i.e., the creation of policies or other interventions that result in emissions reductions over and above what would have occurred independently. This shift is stronger among firms with active (and more ambitious) Scope 2 emissions targets, consistent with targets increasing the demand for reported reductions when near-term physical abatement is costly. Collectively, our evidence highlights how emissions reporting protocols can shape firms’ efforts to achieve emission-reduction goals, with direct implications for ongoing Scope 2 standard-setting debates as well as the mapping of firms’ contractual claims to their real-world emissions impact.

Our study provides timely evidence as policymakers seek feedback on proposed updates to existing Scope 2 guidance. The current market-based approach to reporting emissions offers flexibility in measuring progress, recognizing that differences in business models, geographic locations, and operational barriers require alternative approaches to reducing emissions. This flexibility comes at the cost of weakening the physical and temporal link to actual energy consumption. Indeed, firms can choose where, when, and what type of renewable energy source to support without altering their actual energy use. Nonetheless, the unbundling of the renewable energy attribute increases the supply of unbundled instruments, creating a low-cost option for reporting lower emissions that sacrifices additionality in service to flexibility.

Evaluating progress toward decarbonization targets requires attention not only to whether firms report reductions, but also to the mechanisms through which those reductions are achieved. Ultimately, our findings suggest that the current approach to reporting Scope 2 emissions plays a role in determining how firms choose to decarbonize and, in turn, shapes global efforts to mitigate climate change. Indeed, given the AI-driven surge in Scope 2 emissions, our evidence calls

attention to the important role that carbon accounting plays in determining how tech companies account for data center emissions, as well as whether and how tech firms hinder genuine progress in reducing global energy consumption.

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Appendix A: Variable definitions.

DEPENDENT VARIABLES	
<i>RenEnergyInstr_dummy</i>	An indicator variable equal to one if a firm purchases renewable energy instruments to claim low-carbon electricity, heat, steam, or cooling.
<i>Ln(RenEnergyInstr)</i>	The natural logarithm of the megawatt-hours (MWh) of low-carbon electricity, heat, steam, or cooling.
<i>sinh-1(RenEnergyInstr)</i>	The inverse hyperbolic sine (IHS) transformation of the megawatt-hours (MWh) of low-carbon electricity, heat, steam, or cooling, scaled by the firm's revenue.
<i>sinh-1(BundledREI)</i>	The IHS transformation of the MWh of low-carbon electricity, heat, steam, or cooling obtained through bundled contracts for renewable energy, scaled by the firm's revenue.
<i>sinh-1(UnbundledREI)</i>	The IHS transformation of the MWh of low-carbon electricity, heat, steam, or cooling obtained through unbundled purchases, scaled by the firm's revenue.
<i>NonRenewEnergy</i>	The percentage of nonrenewable energy consumption defined as total energy consumption minus purchased electricity, steam, heat, or cooling derived from renewable sources scaled by total revenue.
<i>RenewEnergy</i>	The percentage of renewable energy consumption defined as purchased electricity, steam, heat, or cooling derived from renewable sources scaled by total revenue.
<i>Scope1</i>	Scope 1 emissions scaled by the firm's revenue and divided by 100.
INDEPENDENT VARIABLES	
<i>HighScope2</i>	Firms with above-industry-median Scope 2 emissions intensity in 2014.
<i>Post</i>	Indicator variable equal to one for firm-years ≥ 2015 .
CONTROL VARIABLES	
<i>Cash</i>	Cash and short-term investments divided by total assets.
<i>Leverage</i>	Sum of short-term and long-term debt divided by total assets.
<i>PPE</i>	Property, plant, and equipment divided by total assets.
<i>ROA</i>	Net income divided by total assets.
<i>Size</i>	The natural logarithm of total assets.
<i>InstOwn</i>	The percentage of institutional ownership by investors who are signatories to the United Nations Principles for Responsible Investment.
CROSS-SECTIONAL VARIABLES	
<i>Target</i>	An indicator variable equal to one if the firm-year has an active Scope 2 target.
<i>AbsoluteTarget</i>	An indicator variable equal to one if the firm-year has an active absolute Scope 2 target.
<i>IntensityTarget</i>	An indicator variable equal to one if the firm-year has an active Scope 2 intensity target.
<i>AmbitiousTarget</i>	An indicator equal to one if the planned carbon reduction scaled by the target horizon falls within the industry-year top quartile.

Appendix B: Types of renewable energy instruments

We classify the market-based Scope 2 renewable energy purchased as bundled and unbundled renewable energy instruments based on their sourcing methods. Panels A and B present the common examples of both types of instruments.

Panel A – Unbundled renewable energy instruments

Freq	Sourcing Method
1	Unbundled Energy Attribute Certificate (EAC) purchase
2	Unbundled energy attribute certificates, International REC Standard (I-RECs)
3	Unbundled energy attribute certificates, Guarantees of Origin (GOs)
4	Green electricity products from an energy supplier (e.g. green tariffs)
5	Renewable Energy Guarantees of Origin (REGO)
6	Default delivered electricity from the grid (e.g. standard product offering by an energy supplier), supported by energy attribute certificates
7	Contract with suppliers or utilities, with a supplier-specific emission rate, not backed by electricity attribute certificates
8	PowerPlus instruments
9	Heat/steam/cooling supply agreement
10	Standard product offering by an energy supplier supported by energy attribute certificates
11	Emission free energy certificates (EFECs) purchase
12	Mixed renewable sources
13	Unbundled energy attribute certificates, International REC Standard (I-RECs)
14	Non-grid connected low carbon electricity not owned by company, no instruments created
15	Unbundled energy attribute certificates, other – please specify: TIGR

Appendix B: Types of renewable energy instruments (concluded)

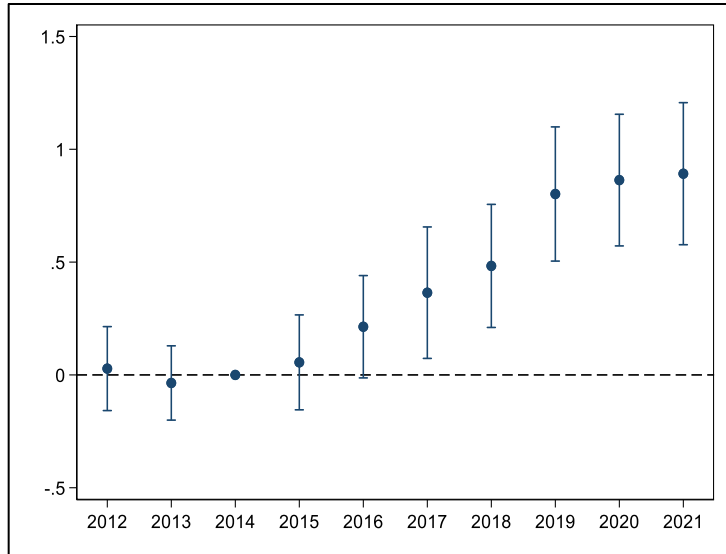
Panel B – Bundled renewable energy instruments

Freq	Sourcing Method
1	Off-grid energy consumption from an on-site installation or through a direct line to an off-site generator owned by another company
2	Power purchase agreement (PPA) with a grid-connected generator with energy attribute certificates
3	Direct procurement from an off-site grid-connected generator e.g. Power purchase agreement (PPA)
4	Power purchase agreement (PPA) with on-site/off-site generator owned by a third party with no grid transfers (direct line)
5	Non-grid connected low carbon heat, steam or cooling, generation owned by company
6	Off-grid energy consumption from an onsite installation or through a direct line to an off-site generator
7	Grid connected low carbon electricity generation owned by company, instruments created and retired by company
8	Direct procurement contract with a grid connected generator or Power Purchase Agreement (PPA), supported by energy attribute certificates
9	Direct line to an off-site generator owned by a third party with no grid transfers
10	Off-grid energy consumption from an on-site installation or through a direct line to an off-site generator owned by another company
11	Renewable On-site self-generation
12	Off-grid energy consumption from an on-site installation or through a direct line to an off-site generator
13	Solar PV installations
14	Renewable electricity self-generation owned by company
15	On-site wind turbine facilities

Figure 1: Parallel trends and dynamic effects

This figure presents event-time estimates from a dynamic difference-in-differences regression to assess the validity of the parallel-trends assumption. The plotted coefficients represent the year-by-year differences in renewable energy instrument adoption between high emitters and low emitters, relative to the base year 2014. Panel A reports the results for total renewable energy instruments, and Panel B reports the results for unbundled renewable energy instruments. All regressions include the full set of control variables and fixed effects as specified in Equation (1). Error bars indicate 90% confidence intervals based on standard errors clustered at the firm level. All continuous variables are winsorized at the 1st and 99th percentile.

Panel A – Total renewable energy instruments



Panel B – Unbundled renewable energy instruments

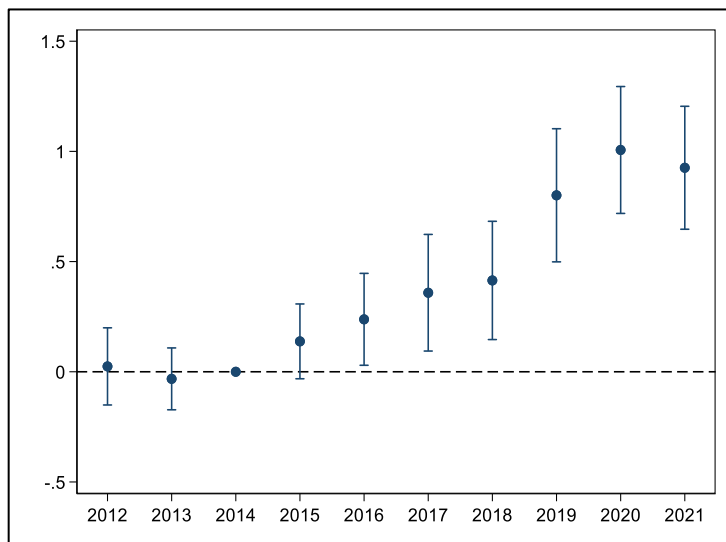


Table 1: Descriptive statistics

This table presents descriptives for the sample, spanning the years 2012 to 2021. Panel A presents the distribution of the sample firms by country. Panel B shows the industry distribution of the sample and reports the sample coverage as a percentage of all Compustat firms. Panel C provides summary statistics for the main variables used in the analysis. Appendix A provides variable definitions. All continuous variables are winsorized at the 1st and 99th percentile.

Panel A – Sample distribution by country

Country	Firm-years	% of sample	Country	Firm-years	% of sample
Australia	119	1.6%	Mexico	49	0.7%
Austria	42	0.6%	Netherlands	69	0.9%
Belgium	51	0.7%	New Zealand	29	0.4%
Brazil	140	1.9%	Norway	117	1.6%
Canada	383	5.1%	Poland	8	0.1%
Chile	13	0.2%	Portugal	58	0.8%
China	47	0.6%	Singapore	20	0.3%
Colombia	14	0.2%	South Africa	277	3.7%
Denmark	71	0.9%	South Korea	258	3.4%
Finland	193	2.6%	Spain	224	3.0%
France	319	4.2%	Sweden	151	2.0%
Germany	275	3.7%	Switzerland	194	2.6%
Hong Kong	48	0.6%	Taiwan	188	2.5%
India	135	1.8%	Thailand	19	0.3%
Ireland	30	0.4%	Turkey	84	1.1%
Israel	18	0.2%	United Kingdom	726	9.6%
Italy	162	2.2%	United States	1,946	25.8%
Japan	1,055	14.0%	Total	7,532	100.0%

Table 1: Descriptive statistics (continued)**Panel B – Industry distribution**

Fama-French 12 Industries	Frequency	% of Sample	% of Compustat Population
Consumer NonDurables	688	9.13	1.99
Consumer Durables	273	3.62	2.03
Manufacturing	1,198	15.91	2.17
Oil, Gas, and Coal Extraction and Production	279	3.70	2.63
Chemicals and Allied Products	515	6.84	3.38
Business Equipment	1,033	13.71	1.93
Telephone and Television Transmission	334	4.43	5.19
Utilities	406	5.39	4.90
Wholesale, Retail, and Some Services	549	7.29	1.76
Healthcare, Medical Equipment, and Drugs	416	5.52	1.62
Finance	332	4.41	2.19
Other	1,509	20.03	2.39
Total	7,532	100	

Table 1: Descriptive statistics (continued)**Panel C – Summary statistics**

VARIABLES	# Obs.	Mean	Std. Dev.	P25	P50	P75
<i>sinh-1(RenEnergyInstr)</i>	7,532	1.402	1.881	0	0.091	2.722
<i>Ln(RenEnergyInstr)</i>	4,950	10.370	2.908	8.666	10.770	12.410
<i>RenEnergyInstr_Dummy</i>	7,532	0.670	0.470	0	1	1
<i>sinh-1(BundledREI)</i>	7,532	0.443	1.201	0	0	0.012
<i>sinh-1(UnbundledREI)</i>	7,532	1.096	1.702	0	0	2.086
<i>Cash</i>	7,532	0.119	0.099	0.048	0.0908	0.160
<i>ROA</i>	7,532	0.047	0.056	0.019	0.0423	0.073
<i>Leverage</i>	7,532	0.264	0.160	0.147	0.251	0.364
<i>Size</i>	7,532	10.560	2.520	8.788	10.10	12.060
<i>PPE</i>	7,532	0.284	0.217	0.111	0.240	0.406
<i>InstOwn</i>	7,532	11.960	13.110	1.924	7.629	16.880
<i>Target</i>	7,532	0.377	0.485	0	0	1
<i>AbsoluteTarget</i>	7,532	0.254	0.435	0	0	1
<i>IntensityTarget</i>	7,532	0.178	0.383	0	0	0
<i>TargetReduction_Horizon</i>	2,631	3.637	2.802	2	2.857	4.286

Table 2: Baseline results

This table reports estimates from the difference-in-differences regressions for the sample period from 2012 to 2021, examining the effect of the 2015 GHG Protocol Scope 2 Guidance on firms' adoption of renewable energy instruments. *HighScope2* is an indicator equal to one for firms if their Scope 2 intensity is above the industry-median in 2014. *Post* is an indicator equal to one for years 2015 and onward. The dependent variable in columns (1)–(3) is *RenEnergyInstr_dummy*. Columns (4)–(6) use $\ln(\text{RenEnergyInstr})$. Columns (7)–(9) use $\sinh^{-1}(\text{RenEnergyInstr})$. See Appendix A for definitions of other variables. All continuous variables are winsorized at the 1st and 99th percentile. t-statistics are reported in parentheses, based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	<i>RenEnergyInstr_dummy</i>			$\ln(\text{RenEnergyInstr})$			$\sinh^{-1}(\text{RenEnergyInstr})$		
HighScope2 × Post	0.0523*	0.0565*	0.0734**	0.5466**	0.5653***	0.8125***	0.3506***	0.3500***	0.4027***
	(1.69)	(1.81)	(2.25)	(2.51)	(2.60)	(3.54)	(2.93)	(2.92)	(3.17)
Cash		0.3180**	0.3330**		-0.7192	-0.3147		0.3231	0.4447
		(2.40)	(2.51)		(-0.90)	(-0.39)		(0.83)	(1.13)
ROA		-0.1028	-0.0582		-0.7313	-1.1221		-0.6431	-0.7011
		(-0.72)	(-0.39)		(-0.85)	(-1.20)		(-1.29)	(-1.44)
Leverage		-0.0805	-0.0011		0.5733	0.8832		0.2068	0.4386
		(-0.83)	(-0.01)		(1.01)	(1.46)		(0.64)	(1.33)
Size		0.0666**	0.0613**		0.3226*	0.4003***		-0.1565	-0.1948**
		(2.52)	(2.19)		(1.90)	(2.77)		(-1.56)	(-2.06)
PPE		0.1120	0.1328		0.5384	-0.1992		0.1685	0.1870
		(0.73)	(0.82)		(0.56)	(-0.18)		(0.33)	(0.34)
InstOwn		0.0016	0.0009		-0.0004	-0.0010		-0.0027	-0.0037
		(0.93)	(0.53)		(-0.04)	(-0.09)		(-0.48)	(-0.62)
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Industry-Year FE	N	N	Y	N	N	Y	N	N	Y
Observations	7,532	7,532	7,507	4,950	4,950	4,896	7,532	7,532	7,507
R-squared	0.623	0.625	0.651	0.773	0.774	0.796	0.754	0.754	0.775

Table 3: Bundled vs. unbundled renewable energy certificates

This table presents estimates from the difference-in-differences regression for the sample period from 2012 to 2021, comparing the effects of the 2015 GHG Protocol Scope 2 Guidance on firms' use of unbundled and bundled renewable energy certificates. *HighScope2* is an indicator equal to one for firms with above-median Scope 2 intensity within their industry in 2014. *Post* is an indicator equal to one for years 2015 and onward. The dependent variable in columns (1)–(3) is $\sinh^{-1}$ (*UnbundledREI*), the inverse hyperbolic sine (IHS) transformation of the megawatt-hours (MWh) of low-carbon energy purchased through unbundled renewable energy instruments, scaled by firm revenue. Columns (4)–(6) use $\sinh^{-1}$ (*BundledREI*), the IHS transformation of the MWh of low-carbon energy obtained through bundled renewable energy instruments, scaled by firm revenue. See Appendix A for definitions of other variables. All continuous variables are winsorized at the 1st and 99th percentile. t-statistics are reported in parentheses, based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	$\sinh^{-1}$ (<i>UnbundledREI</i>)			$\sinh^{-1}$ (<i>BundledREI</i>)		
HighScope2 × Post	0.3769*** (3.46)	0.3700*** (3.41)	0.4316*** (3.98)	-0.0132 (-0.13)	-0.0094 (-0.09)	-0.0078 (-0.08)
Cash		-0.3711 (-0.91)	-0.2663 (-0.68)		0.7190** (2.08)	0.7328** (2.12)
ROA		- 1.2395*** (-2.62)	-0.8823* (-1.87)		0.7299* (1.73)	0.4688 (1.26)
Leverage		0.4273 (1.28)	0.6058* (1.86)		0.0302 (0.12)	0.1468 (0.57)
Size		-0.1779 (-1.60)	-0.2076** (-2.16)		0.0028 (0.04)	-0.0147 (-0.18)
PPE		-0.4851 (-0.93)	-0.2854 (-0.52)		0.7367 (1.52)	0.4879 (0.99)
InstOwn		-0.0002 (-0.04)	-0.0017 (-0.30)		-0.0025 (-0.52)	-0.0005 (-0.11)
Firm FE	Y	Y	Y	Y	Y	Y
Country-Year FE	Y	Y	Y	Y	Y	Y
Industry-Year FE	N	N	Y	N	N	Y
Observations	7,532	7,532	7,507	7,532	7,532	7,507
R-squared	0.708	0.709	0.734	0.573	0.574	0.616

Table 4: Emission Reduction Targets

This table presents difference-in-differences-in-differences regression results based on emission reduction target characteristics for the sample period from 2012 to 2021. *HighScope2* is an indicator equal to one for firms with above-median Scope 2 intensity within their industry in 2014. *Post* is an indicator equal to one for years 2015 and onward. In Panel A, *Target* is an indicator equal to one for firms with active Scope 2 carbon emission targets. In Panel B, *TargetType* is *AbsoluteTarget* (*IntensityTarget*) in Columns (1) through (3) (Columns (4) through (6)), an indicator variable equal to one if the firm has an absolute (intensity) Scope 2 emissions target in the respective year, and zero otherwise. In Panel C, *AmbitiousTarget* equals one if a firm's target ambitiousness falls within the top quartile of its industry in a given year, and zero otherwise. Target ambitiousness is measured as the ratio of the planned carbon reduction to the target horizon. See Appendix A for definitions of other variables. All continuous variables are winsorized at the 1st and 99th percentile. t-statistics are reported in parentheses, based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Panel A: Existence of Scope 2 targets

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	<i>sinh-1 (RenEnergyInstr)</i>			<i>sinh-1 (UnbundledREI)</i>			<i>sinh-1 (BundledREI)</i>		
HighScope2×Post×Target	0.4778*** (2.86)	0.4787*** (2.87)	0.5213*** (3.08)	0.5031*** (3.03)	0.4995*** (3.01)	0.5273*** (3.12)	0.1185 (0.78)	0.1268 (0.84)	0.1107 (0.72)
HighScope2×Post	0.1946 (1.45)	0.1942 (1.44)	0.2396* (1.70)	0.2154* (1.95)	0.2102* (1.90)	0.2670** (2.37)	-0.0517 (-0.46)	-0.0507 (-0.45)	-0.0392 (-0.35)
Post×Target	-0.0403 (-0.30)	-0.0387 (-0.29)	-0.0618 (-0.45)	-0.0423 (-0.33)	-0.0313 (-0.24)	-0.0951 (-0.74)	0.0321 (0.27)	0.0209 (0.18)	0.0749 (0.63)
HighScope2×Target	0.2057** (2.35)	0.2062** (2.39)	0.1426 (1.53)	0.0486 (0.52)	0.0540 (0.58)	0.0400 (0.39)	0.2275** (2.46)	0.2210** (2.39)	0.1639* (1.68)
Target	-0.1021 (-1.42)	-0.1030 (-1.45)	-0.0663 (-0.86)	-0.0134 (-0.20)	-0.0219 (-0.32)	0.0213 (0.30)	-0.0963 (-1.51)	-0.0863 (-1.36)	-0.0803 (-1.21)
Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Industry-Year FE	N	N	Y	N	N	Y	N	N	Y
Observations	7,532	7,532	7,507	7,532	7,532	7,507	7,532	7,532	7,507
R-squared	0.757	0.758	0.778	0.711	0.713	0.737	0.575	0.577	0.618

Table 4: Emission Reduction Targets (continued)

Panel B – Target type

	(1)	(2)	(3)	(4)	(5)	(6)
Target Type	Absolute Target			Intensity Target		
VARIABLES	<i>sinh</i> −1 (UnbundledREI)			<i>sinh</i> −1 (UnbundledREI)		
HighScope2×Post×TargetType	0.8087*** (4.19)	0.7975*** (4.14)	0.8297*** (4.20)	0.0167 (0.08)	0.0194 (0.09)	0.0222 (0.11)
HighScope2×Post	0.1984* (1.86)	0.1946* (1.82)	0.2556** (2.38)	0.3891*** (3.43)	0.3823*** (3.37)	0.4403*** (3.84)
TargetType×Post	-0.1237 (-1.14)	-0.1200 (-1.11)	-0.1305 (-1.15)	0.1168 (1.02)	0.1193 (1.06)	0.0832 (0.70)
TargetType×HighScope2	-0.2119 (-1.40)	-0.2035 (-1.34)	-0.2675* (-1.73)	0.1460 (0.86)	0.1536 (0.91)	0.1043 (0.63)
TargetType	0.0759 (0.95)	0.0761 (0.95)	0.1262 (1.51)	-0.0867 (-1.08)	-0.0968 (-1.22)	-0.0528 (-0.63)
Controls	No	Yes	Yes	No	Yes	Yes
Firm FE	Y	Y	Y	Y	Y	Y
Country-Year FE	Y	Y	Y	Y	Y	Y
Industry-Year FE	N	N	Y	N	N	Y
Observations	7,532	7,532	7,507	7,532	7,532	7,507
R-squared	0.574	0.575	0.617	0.708	0.710	0.735

Table 4: Emission Reduction Targets (continued)
Panel C – Target Ambition

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
VARIABLES	<i>sinh-1 (RenEnergyInstr)</i>	<i>sinh-1 (UnbundledREI)</i>	<i>sinh-1 (BundledREI)</i>						
HighScope2×Post×AmbitiousTarget	0.6766** (2.07)	0.6671** (2.08)	0.6398* (1.74)	0.7555* (1.85)	0.7700* (1.91)	0.8247** (2.05)	-0.1279 (-0.41)	-0.1527 (-0.50)	-0.2738 (-0.75)
HighScope2×Post	0.3250 (1.41)	0.3432 (1.53)	0.3552 (1.34)	0.4902* (1.93)	0.4933** (1.97)	0.5470* (1.94)	-0.0518 (-0.25)	-0.0408 (-0.20)	-0.1460 (-0.60)
AmbitiousTarget×Post	-0.3391* (-1.86)	-0.3351* (-1.87)	-0.3423 (-1.58)	-0.3659 (-1.47)	-0.3839 (-1.56)	-0.4665* (-1.85)	-0.0291 (-0.15)	-0.0099 (-0.05)	0.1379 (0.58)
AmbitiousTarget×HighScope2	-0.4741 (-1.61)	-0.4598 (-1.56)	-0.4153 (-1.23)	-0.5751 (-1.59)	-0.5636 (-1.56)	-0.6380* (-1.72)	0.2271 (0.88)	0.2325 (0.91)	0.3667 (1.23)
AmbitiousTarget	0.3279** (2.10)	0.3223** (2.07)	0.3660* (1.94)	0.3918* (1.91)	0.3963* (1.94)	0.5344** (2.56)	-0.0777 (-0.48)	-0.0868 (-0.53)	-0.1878 (-0.99)
Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Industry-Year FE	N	N	Y	N	N	Y	N	N	Y
Observations	2,631	2,631	2,555	2,631	2,631	2,555	2,631	2,631	2,555
R-squared	0.879	0.880	0.906	0.822	0.823	0.861	0.702	0.703	0.762

Table 5: Renewable energy transition

The sample period spans the years 2017 to 2021. *HighScope2* is an indicator equal to one for firms with above-median Scope 2 intensity within their industry in 2014. Panel A reports results for renewable energy consumption as dependent variable in regressions with a *HighScope2* firm indicator (Columns (1) through (3)) plus control variables and fixed effects. The dependent variable, *RenewEnergy*, measures the percentage of purchased electricity, steam, heat, or cooling derived from renewable sources. Columns (4) through (6) of Panel A further interact *HighScope2* with *Time*, defined as distance of the firm-year observation from the first year of the sample period, 2017. Panel B reports results for regressions of firms' nonrenewable energy consumption (*NonRenewEnergy*) on control variables, fixed effects, and the *HighScope2* indicator interacted with *High*, defined as an indicator equal to one for firms with above-median purchases of total renewable energy instruments (*HighTotalREI*) in Columns (1) and (2); unbundled renewable energy instruments (*HighUnbundledREI*) in Columns (3) and (4); bundled renewable energy instruments (*HighBundledREI*) in Columns (5) and (6). See Appendix A for definitions of other variables. All continuous variables are winsorized at the 1st and 99th percentile. t-statistics are reported in parentheses, based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Panel A. Renewable Energy Consumption

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)
	RenewEnergy					
HighScope2	-6.9879*** (-3.38)	-6.5424*** (-3.27)	-5.6065*** (-3.12)			
HighScope2 × Time				0.4520 (0.84)	0.4673 (0.88)	0.8393 (1.60)
Cash		34.6206*** (3.37)	24.1655** (2.24)		3.6879 (0.57)	0.3212 (0.05)
ROA		28.6970** (2.56)	27.5868** (2.39)		-3.5050 (-0.50)	-6.2702 (-0.87)
Leverage		4.5702 (0.85)	4.2363 (0.73)		0.7008 (0.15)	3.7098 (0.80)
Size		0.4922 (0.84)	-0.0966 (-0.15)		1.2506 (0.88)	1.1254 (0.75)
PPE		-5.1763 (-1.43)	4.5282 (0.93)		13.8169* (1.81)	3.2051 (0.40)
InstOwn		-0.0551 (-0.97)	-0.0434 (-0.71)		-0.0087 (-0.15)	-0.0666 (-1.08)
Firm FE	N	N	N	Y	Y	Y
Country-Year FE	Y	Y	Y	Y	Y	Y
Industry-Year FE	N	N	Y	N	N	Y
Observations	3,105	3,105	3,093	2,982	2,982	2,972
R-squared	0.208	0.241	0.391	0.891	0.892	0.904

Table 5: Renewable energy transition (continued)
Panel B – Nonrenewable energy consumption

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>HighTotalREI</i>		<i>HighUnbundledREI</i>		<i>HighBundledREI</i>	
VARIABLES	<i>NonrenewEnergy</i>					
HighScope2 × High	-0.1646 (-0.42)	-0.1361 (-0.31)	-0.1277 (-0.44)	-0.0363 (-0.11)	-0.5890** (-2.00)	-0.6172** (-2.37)
HighREI	-0.4990* (-1.65)	-0.3086 (-0.86)	-0.2081* (-1.68)	-0.1516 (-0.79)	-0.0153 (-0.16)	0.0196 (0.17)
Cash	-1.7181 (-1.11)	-2.6988* (-1.76)	-1.8963 (-1.23)	-2.7829* (-1.83)	-1.8220 (-1.19)	-2.6925* (-1.82)
ROA	-1.4209 (-0.52)	0.5865 (0.25)	-1.1803 (-0.43)	0.7230 (0.31)	-0.9912 (-0.36)	0.8606 (0.37)
Leverage	5.2688** (2.35)	4.3652** (2.50)	5.2781** (2.34)	4.3499** (2.48)	5.0987** (2.29)	4.2133** (2.44)
Size	-1.4742 (-1.26)	-1.2780 (-1.63)	-1.4572 (-1.23)	-1.2503 (-1.59)	-1.4331 (-1.23)	-1.2284 (-1.59)
PPE	-1.4775 (-0.41)	-3.3247 (-1.20)	-1.5929 (-0.44)	-3.3724 (-1.20)	-1.4471 (-0.40)	-3.1798 (-1.15)
InstOwn	0.0117 (0.39)	-0.0028 (-0.12)	0.0117 (0.38)	-0.0026 (-0.12)	0.0107 (0.35)	-0.0031 (-0.14)
Firm FE	Y	Y	Y	Y	Y	Y
Country-Year FE	Y	Y	Y	Y	Y	Y
Industry-Year FE	N	Y	N	Y	N	Y
Observations	2,992	2,982	2,992	2,982	2,992	2,982
R-squared	0.846	0.873	0.846	0.872	0.846	0.873

Table 6: Scope 2 assurance sample

This table reports estimates from the difference-in-differences regression specified in Equation (1), examining the effect of the 2015 GHG Protocol Scope 2 Guidance on firms' adoption of renewable energy instruments. The sample period spans the years 2012 to 2021. The sample is restricted to firm-year observations that include assurance or verification of Scope 2 emissions. *HighScope2* is an indicator equal to one for firms with above-median Scope 2 intensity within their industry in 2014. *Post* is an indicator equal to one for years 2015 and onward. The dependent variable in columns (1)–(3) is $\sinh^{-1}(\text{RenEnergyInstr})$, the inverse hyperbolic sine (IHS) transformation of the megawatt-hours (MWh) of low-carbon energy usage scaled by firm revenue. Columns (4)–(6) use $\sinh^{-1}(\text{UnbundledREI})$, the IHS transformation of the MWh of low-carbon energy purchased through unbundled contracts, scaled by firm revenue. Columns (7)–(9) use $\sinh^{-1}(\text{BundledREI})$, the IHS transformation of the MWh of low-carbon energy obtained through bundled contracts, scaled by firm revenue. See Appendix A for definitions of other variables. All continuous variables are winsorized at the 1st and 99th percentile. t-statistics are reported in parentheses, based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	$\sinh^{-1}(\text{RenEnergyInstr})$			$\sinh^{-1}(\text{UnbundledREI})$			$\sinh^{-1}(\text{BundledREI})$		
HighScope2×Post	0.6360*** (3.50)	0.6273*** (3.47)	0.7340*** (3.82)	0.6686*** (3.28)	0.6528*** (3.24)	0.7768*** (3.78)	0.0158 (0.08)	0.0219 (0.12)	-0.0107 (-0.05)
Cash		-0.4845 (-0.92)	-0.3097 (-0.55)		-0.4600 (-0.77)	-0.4169 (-0.68)		-0.0229 (-0.05)	0.0198 (0.04)
ROA		-1.4617* (-1.89)	-1.5727** (-2.28)		-2.0059** (-2.55)	-1.7415*** (-2.59)		0.9306 (1.62)	0.9533 (1.49)
Leverage		0.2336 (0.51)	0.1987 (0.45)		0.5181 (1.01)	0.5322 (1.12)		0.1530 (0.38)	0.2035 (0.46)
Size		-0.2203* (-1.94)	-0.1342 (-1.16)		-0.3081** (-2.23)	-0.3079** (-2.53)		0.0957 (0.88)	0.1666 (1.32)
PPE		0.8471 (1.27)	0.8543 (1.11)		0.1155 (0.15)	0.2890 (0.35)		1.0743 (1.51)	0.6733 (0.87)
InstOwn		-0.0034 (-0.50)	-0.0076 (-1.03)		-0.0012 (-0.15)	-0.0034 (-0.42)		-0.0057 (-0.82)	-0.0059 (-0.80)
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country-Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Industry-Year FE	N	N	Y	N	N	Y	N	N	Y
Observations	4,086	4,086	4,028	4,086	4,086	4,028	4,086	4,086	4,028
R-squared	0.834	0.835	0.858	0.774	0.776	0.814	0.661	0.662	0.710

Table 7: Scope 1 emissions

This table reports estimates from a difference-in-differences regression examining the effect of the 2015 GHG Protocol Scope 2 Guidance on firms' Scope 1 emissions. The sample period spans the years 2012 to 2021. The dependent variable, *Scope1*, is the Scope 1 emissions scaled by the firm's revenue and divided by 100. *HighScope2* is an indicator equal to one for firms with above-median Scope 2 intensity within their industry in 2014. *Post* is an indicator equal to one for years 2015 and onward. See Appendix A for definitions of other variables. All continuous variables are winsorized at the 1st and 99th percentile. t-statistics are reported in parentheses, based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

	(1)	(2)	(3)
VARIABLES	<i>Scope1</i>		
HighScope2 × Post	-0.0269 (-0.21)	-0.0561 (-0.42)	-0.0733 (-0.51)
Cash		-0.4696* (-1.82)	-0.5998** (-2.19)
ROA		- (-3.30)	-1.3056* (-1.91)
Leverage		-0.0059 (-0.02)	-0.1369 (-0.47)
Size		-0.7663** (-2.24)	-0.7000** (-2.06)
PPE		-0.0970 (-0.22)	-0.2696 (-0.61)
InstOwn		-0.0015 (-0.44)	-0.0049 (-1.24)
Firm FE	Y	Y	Y
Country-Year FE	Y	Y	Y
Industry-Year FE	N	N	Y
Observations	7,524	7,524	7,499
R-squared	0.973	0.974	0.976

Table 8: Exclusion of firms headquartered in the European Union

This table reports estimates from difference-in-differences regressions, examining the effect of the 2015 GHG Protocol Scope 2 Guidance on firms' adoption of renewable energy instruments. The sample period spans the years 2012 to 2021. The sample excludes observations for firms headquartered in the European Union, which are subject to the Non-Financial Reporting Directive (NFRD). The dependent variable is $\sinh^{-1}(\text{RenEnergyInstr})$, the inverse hyperbolic sine (IHS) transformation of the MWh of low-carbon energy usage scaled by firm revenue. *HighScope2* is an indicator equal to one for firms with above-median Scope 2 intensity within their industry in 2014. *Post* is an indicator equal to one for years 2015 and onward. See Appendix A for definitions of other variables. All continuous variables are winsorized at the 1st and 99th percentile. t-statistics are reported in parentheses, based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

	(1)	(2)	(3)
VARIABLES	$\sinh^{-1}(\text{RenEnergyInstr})$		
HighScope2 $\times$ Post	0.3047** (2.37)	0.3124** (2.43)	0.3751*** (2.76)
Cash		0.5924 (1.45)	0.7974* (1.89)
ROA		-0.6722 (-1.24)	-0.7328 (-1.44)
Leverage		0.1880 (0.53)	0.3306 (0.95)
Size		-0.1514 (-1.38)	-0.1946* (-1.95)
PPE		0.4327 (0.78)	0.6516 (1.10)
InstOwn		-0.0072 (-1.25)	-0.0098 (-1.60)
Firm FE	Y	Y	Y
Country-Year FE	Y	Y	Y
Industry-Year FE	N	N	Y
Observations	5,879	5,879	5,853
R-squared	0.735	0.736	0.765

Table 9: Disclosure Gap

This table reports results examining whether high emitters exhibit a larger Scope 2 emissions disclosure gap compared to low emitters. The sample is restricted to firm-year observations from 2015 to 2021. The dependent variable, *DisclosureGap*, is defined as the difference between location-based and market-based Scope 2 emissions, scaled by firm revenue. *HighScope2* is an indicator equal to one for firms with above-median Scope 2 intensity within their industry in 2014. See Appendix A for definitions of other variables. All continuous variables are winsorized at the 1st and 99th percentile. t-statistics are reported in parentheses, based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

	(1)	(2)	(3)
VARIABLES	<i>DisclosureGap</i>		
HighScope2	4.0796*** (4.26)	3.3290*** (3.81)	3.7385*** (4.17)
Cash		7.3693** (2.15)	3.4798 (0.94)
ROA		-10.3410 (-1.52)	-2.4957 (-0.33)
Leverage		0.6050 (0.26)	1.1452 (0.45)
Size		-0.1433 (-0.53)	-0.3239 (-1.09)
PPE		12.1091*** (4.85)	6.5180** (2.29)
InstOwn		0.0397 (1.39)	0.0358 (1.13)
Country-Year FE	Y	Y	Y
Industry-Year FE	N	N	Y
Observations	4,730	4,730	4,714
R-squared	0.098	0.127	0.219

Table 10: Robustness

This table reports robustness tests for the difference-in-differences regression, which examines the effect of the 2015 GHG Protocol Scope 2 Guidance on firms' adoption of renewable energy instruments. Panel A includes additional interaction terms between the *Post* indicator and all control variables to account for potential differential trends. Panel B presents results using an entropy-balanced sample, by entropy balancing the treatment (*HighScope2* = 1) and control firms (*HighScope2* = 0) in 2014 on the first and second moment of the included control variables. Panel C reports estimates from a Poisson regression as an alternative modeling approach. See Appendix A for variable definitions. All continuous variables are winsorized at the 1st and 99th percentile. t-statistics are reported in parentheses, based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% two-tailed levels, respectively.

Panel A – Interactive controls

VARIABLES	(1)	(2)	(3)
	<i>sinh-1 (RenEnergyInstr)</i>		
HighScope2 × Post	0.3672*** (3.04)	0.5541*** (5.08)	0.4023*** (3.09)
Cash	0.1300 (0.27)	0.4391 (0.87)	0.2521 (0.52)
ROA	-1.0857 (-1.51)	-1.0436 (-1.46)	-1.1315 (-1.61)
Leverage	-0.0282 (-0.07)	0.1802 (0.45)	0.2605 (0.64)
Size	-0.1764* (-1.69)	-0.1975* (-1.88)	-0.1982** (-2.01)
PPE	0.3418 (0.64)	0.6996 (1.14)	0.1516 (0.25)
InstOwn	0.0015 (0.20)	-0.0038 (-0.51)	-0.0028 (-0.37)
Cash × Post	0.3304 (0.85)	-0.0350 (-0.09)	0.3082 (0.80)
ROA × Post	0.5905 (0.89)	0.4718 (0.62)	0.6161 (0.85)
Leverage × Post	0.3177 (0.99)	0.1933 (0.60)	0.2315 (0.68)
Size × Post	-0.0074 (-0.24)	-0.0296 (-1.50)	0.0012 (0.03)
PPE × Post	-0.2582 (-1.04)	-0.2401 (-0.74)	0.0449 (0.14)
InstOwn × Post	-0.0030 (-0.61)	-0.0028 (-0.52)	-0.0007 (-0.12)
Firm FE	Y	Y	Y
Country-Year FE	N	Y	Y
Industry-Year FE	N	N	Y
Observations	7,559	7,532	7,507
R-squared	0.750	0.755	0.775

Table 10: Robustness (continued)
Panel B – Entropy-balanced sample

	(1)	(2)	(3)
VARIABLES	<i>sinh-1 (RenEnergyInstr)</i>		
HighScope2 × Post	0.5587*** (3.01)	0.5601*** (3.03)	0.5585*** (3.41)
Cash		0.4909 (1.07)	0.4975 (1.05)
ROA		-0.5315 (-0.91)	-0.8689 (-1.46)
Leverage		0.4992 (1.24)	0.6874 (1.61)
Size		-0.0478 (-0.40)	-0.1575 (-1.40)
PPE		-0.1316 (-0.21)	-0.1739 (-0.25)
InstOwn		-0.0059 (-0.80)	-0.0049 (-0.66)
Firm FE	Y	Y	Y
Country-Year FE	Y	Y	Y
Industry-Year FE	N	N	Y
Observations	7,509	7,509	7,484
R-squared	0.720	0.720	0.753

Table 10: Robustness (concluded)
Panel C – Poisson Regression

	(1)	(2)	(3)
VARIABLES	<i>RenEnergyInstr</i>		
HighScope2 × Post	0.6219** (2.53)	0.6704*** (2.81)	0.6344*** (3.32)
Cash		0.1908 (0.25)	0.8627 (1.08)
ROA		-0.0913 (-0.10)	-0.7037 (-0.80)
Leverage		1.2801** (2.17)	1.3481** (2.12)
Size		-0.2872* (-1.88)	- (-3.39)
PPE		0.5210 (0.63)	0.2182 (0.23)
InstOwn		-0.0112 (-1.37)	-0.0098 (-0.95)
Firm FE	Y	Y	Y
Country-Year FE	Y	Y	Y
Industry-Year FE	N	N	Y
Observations	6,614	6,614	6,550